University of Engineering & Technology Peshawar

Programming Fundamentals (C++) Lab

LAB REPORT 5

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LAB TITLE:	**The WHILE LOOPS IN C++**

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SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

The objective of this lab is to understand and implement looping structures in C++. The main goals of this lab are:

- To understand the working of the while loop.
- To learn about nested while loops.

- To implement the do-while loop in C++ programs.
- To perform repetitive calculations using loops.

Theory:

While Loop:

A while loop is a control structure that allows a block of code to execute repeatedly as long as a specified condition remains true.

The condition is checked before executing the loop body. If the condition is true, the statements inside the loop are executed. The loop continues until the condition becomes false.

Syntax:

```
while(condition)
{
    // statements
}
```

Nested While Loop:

A nested while loop is a loop inside another loop. The inner loop executes completely for each iteration of the outer loop.

Syntax:

```
while(condition1)
{
  while(condition2)
  {
    // statements
  }
}
```

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Do-While Loop

The do-while loop is similar to the while loop but with one important difference.

The loop body executes at least once, because the condition is checked after executing the statements.

Syntax:

```
do
{
    // statements
}
while(condition);
```

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Lab Tasks & Programs

Task 1: Display Number Using The While Loop:

Description:

In this task, a C++ program was developed using a while loop to display numbers from 1 to 10. The loop starts with an initial value and continues to execute until the condition becomes false. Each number is printed on a separate line to demonstrate how the while loop repeatedly executes the statements inside the loop.

Code:

```
#include <iostream>

using namespace std;



int main()
```

```
{
    int i = 1; // yahan variable i ko 1 se start kiya gaya hai

    while(i <= 10) // jab tak i 10 se chota ya barabar hai loop chalta rahega
    {
        cout << i << endl; // current number screen par print karega
        i++; // i ki value 1 se barh jayegi (increment)
    }

    return 0; // program end
}
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

Task 2:Display Number Using DO-While Loop:

Description:

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In this task, the same program was implemented using a do-while loop. This loop executes the program statements first and then checks the condition. This ensures that the loop body runs at least once, even if the condition is false.

Code:

```
#include <iostream>

using namespace std;


int main()

{

    int i = 1; // variable i ko 1 se start kiya


    do

    {

        cout << i << endl; // number screen par print hoga

        i++; // har dafa i ki value 1 barh jayegi

    }

    while(i <= 10); // condition check hogi ke i 10 se chota ya barabar
hai ya nahi


    return 0; // program finish

}
```

Output:

```
 1

2
```

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```
3

4

5

6

7

8

9

10
```

Task 3: Sum Of Number Using While-Loop:

Description:

In this task, a C++ program was written to calculate the sum of numbers from 1 to a user-entered positive integer. The program takes input from the user and uses a while loop to repeatedly add numbers until the given number is reached. Finally, the calculated sum is displayed.

Code:

```
##include <iostream>

using namespace std;


int main()

{

    int num, i = 1, sum = 0;

    // num user se number lene ke liye

    // i counting ke liye

    // sum total store karne ke liye
```

```
    cout << "Enter a positive integer: "; // user ko message show

    cin >> num; // user se number input lena


    while(i <= num) // jab tak i user ke number se chota ya barabar hai

    {

        sum = sum + i; // har dafa i ko sum me add karna

        i++; // i ko 1 barhana

    }


    cout << "Sum = " << sum; // final sum display


    return 0; // program end

}
```

Output:

```
 Enter a positive integer: 5

Sum = 15
```

Result:

In this lab, we successfully implemented while loop, nested loop concept, and do-while loop in C++. We learned how loops help execute repetitive tasks efficiently.

Conclusion:

This lab helped us understand the basic concept of looping structures in C++.

- We learned how to use the while loop to execute a set of statements repeatedly based on a condition.

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- We also studied the do-while loop, which ensures that the loop body executes at least once.
- We practiced displaying numbers using loops.
- We learned how loops can be used to perform repetitive calculations, such as finding the sum of numbers.
- Overall, this lab improved our understanding of loop control and program execution in C++.

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